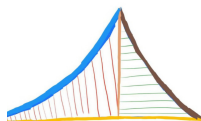


Matrices In Geometry

1



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ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE EC 2026

A. Formulae

B. Solved Questions

1.2.1 The statements (S1), (S2), and (S3) pertain to the scores obtained by students in an exam. The maximum possible marks in the exam is 150.

(S1) The highest score is 100.

(S2) The fourth highest score is 76.

(S3) There are at least four students whose scores are within 25 of each other.

Which of the following options are necessarily correct?

- a) (S1) and (S2) together imply (S3)
- b) (S1) and (S3) together imply (S2)
- c) (S2) and (S3) together imply (S1)
- d) (S1) implies (S3)

1.2.2 Real numbers y , p , and n (all greater than 1) satisfy $(\log_{p^{\frac{1}{n}}} y)(\log_{y^{\frac{1}{n}}} p) = 16$, where the logarithms are taken to the bases $p^{\frac{1}{n}}$ and $y^{\frac{1}{n}}$. The value of n is

- a) 2
- b) 4
- c) 8
- d) 16

1.2.3 Three children P, Q, R and two grown-ups X, Y play a badminton doubles tournament. X and Y are parents to two of the children playing. The child of X is not the same as the child of Y. Exactly one of the children does not have a parent playing in the tournament. The following rules are followed:

- i A parent and his/her child cannot be on the same team.
- ii A match can feature at most one parent and his/her child, that is, a maximum of one parent-child pair can play in a match.

The following matches were played:

	TEAM 1	TEAM 2
MATCH 1	P and X	Q and R
MATCH 2	P and R	X and y
MATCH 3	R and X	Q and Y

Which of the following options is correct?

- a) P does not have any parent playing
- b) Q does not have any parent playing
- c) R does not have any parent playing
- d) X does not have a child playing

1.2.4 Let P_k represent the perimeter of a square with sides of length k . The value of the expression $P_1 + P_2 + P_3 + \dots + P_{10}$ is:

- a) 55
- b) 110
- c) 220
- d) 440

1.2.5 Consider the differential equation $\dot{\mathbf{w}} = A\mathbf{w}$, with $\mathbf{w}(t = 0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. If $w(t) = e^t \mathbf{u}_x + e^{-2t} \mathbf{u}_y$ be the solution to the equation where $\mathbf{u}_x$ and $\mathbf{u}_y$ are unit vectors

along the positive x and y axes respectively, then which of the following options is the correct matrix representing A?

- a) $\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$ b) $\begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$ c) $\begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$ d) $\begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}$

1.2.6 A surface is given by $z^2 = 2x^2 - y^2$ and $\mathbf{n}$ and $-\mathbf{n}$ are the unit vectors are unit normal vectors to the surface at the point $\mathbf{P} = \mathbf{i} + \sqrt{2}\mathbf{k}$ Which of the following vectors can be $\mathbf{n}$, where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ along x, y and z axes, respectively?

- a) $\mathbf{i} - \sqrt{2}\mathbf{k}$ b) $\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{k}$ c) $\sqrt{2}\mathbf{i} - \sqrt{3}\mathbf{k}$ d) $\frac{\sqrt{2}\mathbf{i} - \mathbf{k}}{\sqrt{3}}$

1.2.7 The Laplace Transform of the signal $x(t) = u(t-2) * (tu(t))$ is given by which of the following expressions?

{''*'' represents the convolution operator}

- a) $\frac{e^{-2s}}{s^2(s-2)}$ b) $\frac{e^{-2(s-2)}}{s^3}$ c) $\frac{se^{-2s}}{(s-2)^2}$ d) $\frac{e^{-2s}}{s^3}$

1.2.8 Two analog signals $x_1(t)$ and $x_2(t)$ (t in second), are sampled at a rate $F_s = 40$ Hz, where $x_1(t) = \cos(20\pi t)$, $t \geq 0$, and $x_2(t) = \cos(100\pi t)$, $t \geq 0$. The first ten samples (starting from $t = 0$) are considered for the analysis. Which of the following statements is TRUE?

- a) All of the first three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.
b) All of the last three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.
c) All of the samples of $x_2(t)$ are greater than the corresponding samples of $x_1(t)$.
d) All of the fourth to seventh samples of $x_1(t)$ are equal to the corresponding samples of $x_2(t)$.

1.2.9 The response of a discrete time system $y[n]$ obeys the following relation:

$$y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n].$$

$$\text{The input to the system is } x[n] = \delta[n] - \frac{1}{3}\delta[n-1].$$

Which of the following options is TRUE for $y[n]$?

- a) Stable and causal response
b) Stable and non-causal response
c) Unstable and causal response
d) Unstable and non-causal response

1.2.10 A control system is shown in the Figure. Which option represents the correct transfer function of the system?

- a) $\frac{1}{(s+4)^2}$ b) $\frac{1}{s+4}$ c) $\frac{2}{s+4}$ d) $\frac{1}{(s^2+8s+17)}$

1.2.11 Consider a discrete memoryless source with an alphabet of four source symbols. $s(t)$ is a multi-level $(-1, 0, +1, +2)$ signal representing a long sequence of random symbols from the above source which is generating 104 symbols per second. Which of the following options is the correct value of equivalent Nyquist bandwidth of $s(t)$?

a) 10kHz

b) 64kHz

c) 5kHz

d) 20kHz

1.2.12 Consider the matrix $\mathbf{M} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{pmatrix}$. Which of the following options is/are TRUE if $\det(\mathbf{M}) \neq 0$?

a) $a = -\frac{1}{2}$ and $b = -\frac{1}{2}$ b) $a = \frac{1}{2}$ and $b = \frac{1}{2}$ c) $a = -3$ and $b = 0$ d) $a = \frac{1}{2}$ and $b = -3$

1.2.13 Consider the circuit shown in the Figure, where the input $v_i(t)$ is in Volt. The average power (in mW) dissipated in the load resistance of $1k\Omega$ at the resonant frequency is: (rounded off to two decimal places)

1.2.14 Consider the two series, SA and SB, where

$$S_A = \sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

$$S_B = 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} + \frac{1}{128} + \frac{1}{512} + \dots$$

Which of the following statements is correct for the two given series?

- a) Both S_A and S_B converge.
- b) Neither S_A nor S_B converges.
- c) S_A converges but S_B does not converge.
- d) S_B converges but S_A does not converge.

1.2.15 The continuous time signal $x(t)$ is real, periodic with period T and satisfies the Dirichlet conditions. The Fourier series representation of $x(t) = \sum_{-\infty}^{\infty} a_n e^{j(\frac{2\pi n t}{T})}$ and $x(t)$ satisfies the following: $x(t = \frac{T}{2}) = -x(t)$. For any integer m, which of the following options is correct?

a) $a_{2m} = 0$ b) $a_{2m} = 1$ c) $a_{2m} = a_{2m+1}$ d) $a_{2m} = -1$

1.2.16 Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows Normal distribution with zero mean and unity variance.

Y and Z are defined as, $Y = X + N$ and $Z = X^2 + N$.

Which of the following pairs represents the values of correlation between X and Y and that between X and Z?

- a) $\frac{1}{3}$ and 0
- b) $\frac{1}{3}$ and $\frac{1}{9}$
- c) $\frac{1}{3}$ and $\frac{1}{3}$
- d) 1 and 0

1.2.17 In the given circuit, $L = 1\mu H$ and $C = 1\mu F$. The phasor diagram for IC and IL is also shown. Assume that the phase $(\theta_1 + \theta_2)$ is 90° at a frequency of 159.15 kHz. Among the following options, what is the nearest integer value of $R_C \times R_L$?

a) 0

b) 1

c) 2

d) 10

- 1.2.18 For the control system shown in the Figure, the transfer function of a plant, $G(s) = (s+1)(s+2)$ is connected in cascade with a compensator $c(s) = K(s+\alpha)$, where K and α are positive real valued constants. Which of the following pairs (K, α) represent the correct values for the closed loop system to have poles at $(-3 \pm j\sqrt{5})$?

a) 2,3

b) 3,4

c) 2,4

d) 3,3

- 1.2.19 The state and output equations for a control system are:

$$\dot{x} = \begin{pmatrix} -4 & -1.5 \\ 4 & 0 \end{pmatrix} x + \begin{pmatrix} 2 \\ 0 \end{pmatrix} u$$

$$y = (1.5 \quad 0.625) x$$

Which of the following expressions correctly represents the transfer function $\frac{Y(s)}{u(s)}$ of the system with zero initial conditions?

a) $\frac{3s}{s^2+4s-6}$

b) $\frac{3s+5}{s^2+4s-6}$

c) $\frac{3s+5}{s^2+4s+6}$

d) $\frac{3s}{s^2+4s+6}$

- 1.2.20 The address of the first location of a 256 kilo byte (KB) memory is $(2500)_H$. Choose the correct address of the last location of the memory.

a) $(2FFF)_H$

b) $(124FF)_H$

c) $(424FF)_H$

d) $(324FF)_H$

- 1.2.21 Consider a real signal $x(t)$, $-\infty < t < \infty$, such that $x(t) = 0$ for $t < 0$, $x(t) = 2$ for $0 \leq t < 1$ and $x(t) = 0$ for $t \geq 1$.

Let $E[x(t)] = \int_{-\infty}^{\infty} x(t) 2dt$. Which of the following options correctly represents the ratio, $E[x(t)]/E[3x(-3t+5)]$?

a) 3

b) 1

c) $\frac{1}{3}$ d) $\frac{1}{9}$

- 1.2.22 A QPSK modulated signal from an additive white Gaussian noise (AWGN) channel is received with an $E_b/N_o = 8.4$ dB at the input of a coherent QPSK demodulator. A maximum-likelihood reception method is used in the demodulator. Assume the complimentary error function

$$\text{erfc}(u) \approx [1/(u\sqrt{\pi})] \exp(-u^2)$$

. Which is the nearest bit error rate (BER) at the output of the demodulator?

a) 10^{-3} b) 10^{-4} c) 10^{-5} d) 10^{-6}

- 1.2.23 What is the 10's complement of $(47)_10$?

a) 52

b) 53

c) 54

d) 55

1.2.24 Consider a real baseband signal $x(t) = e^{-2t}$, for t (in seconds) ≥ 0 . If 99% of energy of $x(t)$ lies within B Hz, then which of the following options is TRUE for the value of B ?

a) $B \leq 1\text{kHz}$ b) $63/\pi\text{Hz} \leq B \leq 64/\pi\text{Hz}$ c) $126/\pi\text{Hz} \leq B \leq 128/\pi\text{Hz}$ d) $B \leq 1\text{Hz}$

1.2.25 Consider the discrete time system (S) with input $x[n]$ and output $y[n]$ as shown in the Figure. The two sub-systems represented by their impulse responses $h_1[n]$ and $h_2[n]$ are linear and time invariant. Which of the following statements is necessarily TRUE?

a) S is causal.b) S is linear and time invariant.c) S is linear and time varying.d) S is non-linear.

1.2.26 A Boolean function, $f(x, y, z)$ with x as MSB and z as LSB is realized by 4:1 multiplexer (MUX) with select lines, S_1 and S_0 (S_1 is MSB, S_0 is LSB) and inputs, I_0, I_1, I_2, I_3 as shown in the Figure. Which of the following options is the correct expression of $f(x, y, z)$?

a) $xz + y$ b) $x\bar{y} + z$ c) $xy + \bar{z}$ d) $\bar{x}y + \bar{z}$

1.2.27 A shift-left Shift Register (SR) and a D flip-flop are connected to a synchronized clock as shown in the Figure. Assume that the SR and D flip-flops are initially cleared and the XOR gate has no propagation delay. Which of the following options gives the correct binary representation ($b_7b_6b_5b_4b_3b_2b_1b_0$) of the content of the shift register immediately after the 5th clock transition (positive edge)?

a) 00011111

b) 10111111

c) 00111111

d) 11000011

1.2.28 A small signal source, $V_i(t) = A \cos(105t) + B \sin(107t)$ is applied to a BJT circuit as shown in the Figure. Assume zero source resistance, $V_{BE} = 0.7\text{V}$, $\beta_{dc} = 99$, Early voltage = 100 V and Thermal voltage = 25 mV. Effect of internal parasitic capacitances of the BJT may be neglected. Which expression is the best approximation of the output voltage $V_o(t)$?

a) $-9.1[A \cos(10^5t) + B \sin 10^7t]$ b) $9.1[A \cos(10^5t) - B \sin 10^7t]$ c) $-190.4[A \cos(10^5t) + B \sin 10^7t]$ d) $190.4[A \cos(10^5t) - B \sin 10^7t]$

1.2.29 Consider the four-variable Boolean function, $f(w, x, y, z) = \sum m(0, 2, 5, 7, 8, 10, 13, 14, 15)$ with 'w' as MSB and 'z' as LSB. Which of the following expressions is/are the valid form(s) of $f(w, x, y, z)$?

- a) $\bar{x}\bar{z} + xz + wxy$
- b) $xz + wxy + w\bar{x}\bar{z} + \bar{w}xy(z)$
- c) $\bar{x}\bar{z} + wxy + w\bar{x}\bar{z} + \bar{w}xy\bar{z}$
- d) $\bar{x}\bar{y} + xz + wy\bar{z}$

- 1.2.30 Let $x_1(t) = \cos(2\pi nt)$ and $x_2(t) = 2\sin(4\pi nt)$ represent two sinusoids for a positive integer n and $-\infty < t < \infty$. Which of the following statements about $x_1(t)$ and $x_2(t)$ is/are valid?
- a) $x_1(t)$ and $x_2(t)$ are orthogonal to each other over $0 \leq t < 1/n$.
 - b) $x_1(t)$ and $x_2(t)$ are orthonormal to each other over $0 \leq t < 1/n$.
 - c) $x_1(t)$ is a harmonic of $x_2(t)$.
 - d) $x_1(t)$ and $x_2(t)$ are non-orthogonal to each other over $0 \leq t < 1/(2n)$.
- 1.2.31 Consider the unity negative feedback control system shown in the Figure. The value of gain $K_c(0)$ at which the given system will remain marginally stable is : (Answer in integer)
- 1.2.32 Consider the square region R in the X - Y plane as shown with the dark shading in the Figure. The value of $\int_R (x_2 + y_2 - 1) dx dy$ is . (rounded off to two decimal places)
- 1.2.33 Consider the ideal diodes $D1$ and $D2$ as shown in the Figure with cut-in voltage $V_\gamma = 0$ Volt and $v_i(t)$ is in Volt. The maximum voltage (Volt) of the output $v_o(t)$ is . (rounded off to two decimal places)